

Silvia Arellano Romero

Senior Data Engineer & Architect

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I'm a data engineer with deep experience architecting and building scalable data infrastructure in GCP. I've led the design of end-to-end solutions—from ingestion pipelines to core data models—that support teams across product, operations, finance, and customer experience. I'm strong at understanding business needs and translating them into data products that are reliable, actionable, and easy to work with. Whether replacing third-party tools or setting up self-serve workflows, I focus on building systems that are both technically sound and aligned with how teams actually use data to make decisions.

- SQL
- ETL/ELT
- Data Warehousing
- SQL/NoSQL Management

- BigQuery
- Data Visualization
- Beam/Flink/Spark
- Git

SKILLS

- Cloud Computing
- Python
- Firestore
- IaC/Terraform/Pulumi
- Orchestration/Airflow/Prefect
- SQLX/Dataform/DBT

Playtomic

Lead Data Engineer

- **Reduced pipeline development time from weeks to a few days** through an internal Beam framework and templated CI/CD.
- **Cut streaming pipeline costs by up to 60%** via a GCP cost and monitoring framework adopted across the platform.
- **Enabled subsecond latency in streaming pipelines**, unlocking use cases the previous architecture couldn't support.
- **Developed an agent framework for engineering workflows**, bringing AI assistance into pipeline development, ops, and quality

Global Senior Data Engineer

- **Built Playtomic's first fully automated data ingestion system in GCP**, replacing Hevo with scalable pipelines using Dataflow, Pub/Sub, BigQuery, and Apache Beam.
- **Developed IaC to provision Dataflow jobs**, BigQuery tables, and monitoring resources, making deployments repeatable and efficient.
- **Designed data models** and marts using dimensional modeling and star schemas, ensuring consistency and accuracy across reporting layers.
- **Led integrations with Zendesk, CDPs, Marketing tools** ect to support reverse ETL and improve operational insights.
- **Helped establish a more self-serve data culture** by building standard models, improving documentation, and rolling out curated dashboards.

Siftia Data Company

Data Engineer / Data Product Developer

1. Most Recent Project - BigQuery Data Warehouse Refactoring:

- Led a major refactoring of over 80 scheduled queries into efficient stored procedures in BigQuery.
- Implemented error handling, centralized data cleaning, and managed dynamic unnesting of columns.
- Developed a wrapper procedure, cutting down warehouse refresh time from 1.5 hours to under 18 minutes.

2. Main Responsibilities:

- Architected client-facing reporting solutions using BigQuery, Looker, and GraphQL for intuitive and efficient data visualization.
- Developed a Node.js synchronization repository for post-migration syncs and audits in Firebase, ensuring data consistency and accuracy.
- Managed legacy data migration, involving data preparation in BigQuery and Python-based ETL for importing into Firestore.

MMI Business Consulting

Partner & Database Engineer

- **Geolocation Data Integration and Hashing Process.** Developed a geolocation data hashing and matching process, integrating Google Maps and Bing Maps with SQL Server. Automated the process with Python scripts in SQL Server, greatly improving geospatial data accuracy and efficiency for clients.
- **Co-Creation of Database Health Check Tool.** Co-developed a database Health Check tool to evaluate key metrics like query performance and data safety, facilitating proactive database maintenance.
- **Automation of Marketing Data Processes.** Automated marketing data workflows using ETL packages and T-SQL scripts, enhancing data quality and efficiency for a B2B client.
- **Database Coding Standards Creation.** Led the creation of Database Coding Standards, establishing structured guidelines for SQL procedures and scripts, thereby ensuring coding consistency and efficiency.

October 2025 - Present

April 2024 - Oct 2025

Feb 2023 - April 2024

Oct 2020 - Dec 2023

Worky

Jun 2022 - Feb 2023

Data Engineer/Analyst

- **Development of Data Pipelines and ETL Processes.** Implemented the development of ETL processes using GCP tools (BigQuery, PubSub, Dataflow) to create data pipelines from CloudSQL, Hubspot, Odo, and Google Analytics 4 (G4).
- **Building a Scalable Data Warehouse in BigQuery.** Engineered a BigQuery data warehouse with Type 2 slowly changing dimensions and a galaxy schema, optimizing data structure for detailed tracking and analysis efficiency.
- **Custom Reporting for Stakeholders.** Developed and automated 8 live Looker Studio reports for various divisions, fostering real-time insights and driving the company's shift to a data-driven culture.

MatchCraft

Feb 2021 - Jul 2022

Business Data Engineer

- **Audited digital ads data in MySQL and authored SQL reporting that informed strategy.** Enhanced an internal ad-behavior prediction tool and the 'Estimator' for budget optimization, and led the migration of campaign data from Google Ads and Bing Ads into MatchCraft's platform.

B-Metrics

Sept 2019 - Oct 2020

Biomechanical Data Engineer

- Processed and analyzed biomechanical data for national sports teams using Python, delivering pioneering performance-insight reports for institutional and private clients.

CONSULTING PROJECTS

Thomson Reuters - Data Engineer (Contract)

Feb 2025 - May 2026

- Built streaming and batch data infrastructure using Fivetran, Datastream, and Dataflow, with Pulumi for infrastructure as code. Orchestrated transformations with dbt and Airflow and set up monitoring and alerting, enabling teams to independently own and operate their own pipelines for reliable, self-serve data delivery.

Worky - Data Product Developer (Contractor)

June 2024 - Sept 2024

Developed a custom, end-to-end data product in Power BI. The project involved creating batch ELT pipelines from the production database to BigQuery, performing data modeling and processing, and leveraging DirectQuery capabilities for direct connectivity with Power BI. The solution enabled the creation of a new product sold to new clients by the company. Additionally, I standardized the product to ensure it was replicable for multiple customers, enhancing scalability and broadening the market reach.

GRUPO HOMA Real Estate Developers - Data Solutions Architect (Contractor)

Jan 2022 - Jan 2023

I created an end-to-end, user-friendly data product that emphasizes business automation and data warehousing on GCP, significantly transforming Grupo Homa's decision-making process. I developed a comprehensive BigQuery data warehouse integrating financial tools for centralized data management. Additionally, I designed interactive Looker Studio dashboards for real-time monitoring of construction progress, budget tracking, and sales performance. I also automated property management workflows using Google Cloud Functions, Workflow, Zapier, and Google Apps Script, enhancing operational efficiency.

EDUCATION

University of Chihuahua 2018-2020
Master's Degree in Open Source Software

Emeritus University of Puebla 2013-2018
B.Sc. in Physics

CERTIFICATIONS

On-Course	Professional Data Engineer Certification by Google	Google
Oct 2023	Big Data and Machine Learning Fundamentals	Google
Oct 2023	Modernizing Data Lakes and Data Warehouses with Google Cloud	Google

LANGUAGES

English - Fluent

Spanish - Native